

M.Tech. (II Sem.)

1544

CS 503

**M.Tech. (SECOND SEMESTER)
EXAMINATION, 2024**

COMPUTER SCIENCE

CS 503—Advanced Computer Architecture

TIME ALLOWED : THREE HOURS

Maximum Marks— 60

Attempt six questions in all, selecting two questions from each Section. All questions carry equal marks.

SECTION A

1. (a) What are the main types of processor architecture used in modern computers? 5
(b) What are the key elements and their significance in design of a processor? 5
2. (a) What is a multiprocessor? Explain its architecture. Also differentiate between symmetric and asymmetric multiprocessor. 5
(b) Explain multicomputer architecture. What is the role of virtual shared memory (VSM) and shared virtual memory (SVM) in multicomputers. 5

Turn over

3. (a) What is computer performance? Elaborate the factors that impact the performance of a computer system. 5
- (b) Machine A runs a program with 2 GHz clock in 10 seconds. What clock rate must Machine B run to run the same program in 6 seconds? Also calculate the performances of Machine A and Machine B. 5

SECTION B

4. (a) What are vector processors? Explain the principle of SIMD vector processing. 5
- (b) Explain the role of following in vector processor:—
 - (i) Vector register
 - (ii) Vector functional units
 - (iii) Vector load store units
 - (iv) Scalar registers
 - (v) Vector execution time. 5

5. What are super scalar processors? Explain their architecture. What are their essential characteristics, advantages and disadvantages? How are they different from parallelism? 10

6. Write short notes on the following:—

- (a) Control flow computers
- (b) Data flow computers
- (c) Multivector multiprocessor
- (d) Program partitioning and scheduling. 10

SECTION C

7. Differentiate between the following:—

- (a) Linear and non-linear pipeline processor
 - (b) Instruction and Arithmetic pipeline design. 10
8. (a) What do you understand by pipeline processing? Why are throughput and latency considered as two main metrics to measure the performance of pipeline processing? 5
 - (b) Consider a water bottle packaging plant. Let there be 3 stages that a bottle should pass through—
 - (i) Inserting the Bottle (I) – Stage-1
 - (ii) Filling water in the Bottle (F) – Stage-2
 - (iii) Sealing the bottle (S) – Stage-3
 If each stage takes one minute to complete its operation, what should be the average time taken to manufacture 1 bottle? 5

9. Explain the superscalar and vector pipeline design in detail. 10